

ISAAC (ZAC) ADJEI

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PROFILE

Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky). Builds across the full engineering stack - from PCB-level analogue circuit design and embedded firmware to production IoT platforms and secure full-stack web applications. Seeking internships, industrial placements and professional experiences in embedded systems, hardware-software integration, software engineering, cybersecurity, cloud, AI/ML and data.

EDUCATION

Aston University, Birmingham, United Kingdom

Oct 2024 - Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Electronics 1, Electronics 2, Introductory Mathematics for Engineering, Power Skills

- Developing strong foundations in software engineering, electronics and applied mathematics while building practical project experience across embedded systems, AI/data and full-stack development

Stanmore College, London, United Kingdom

Sep 2022 - Jul 2024

Pearson BTEC Level 3 National Extended Diploma in Engineering · **D*DD (Distinction*, Distinction, Distinction)**

Best and Most Hardworking Student Award

Relevant Modules: Engineering Product Design and Manufacture, Microcontroller Systems for Engineers, Engineering Principles, Specialist Engineering Project, CAD in Engineering, Electronic Devices and Circuits, Calculus to Solve Engineering Problems, Electronic Measurement and Testing of Circuits

- Completed intensive hands-on training in engineering design, testing and documentation, applying microcontroller, electronics and CAD skills through practical coursework and projects

SKILLS

Languages: C, C++, Python, Java, TypeScript, JavaScript, PHP, SQL, MATLAB

Embedded & Hardware: Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, PCB Layout, Circuit Design, Oscilloscope, I2C / UART / SPI, FPGA, VHDL

Web and Frameworks: Next.js, React, Node.js, Tailwind CSS, PHP, MySQL, Apache

AI/ML and Data: TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV, Jupyter Notebooks

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Vercel, Cloudflare, GitHub Actions

Professional: Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Time Management, Adaptability, Attention to Detail

PROJECTS

PHAEMOS: Smart Maintenance Platform

2026, ongoing

C++ (ESP32, Arduino), C (STM32 HAL), MicroPython (Pico 2W), FastAPI, Next.js 15, PostgreSQL 15, Redis 7, scikit-learn, Docker, Prometheus, Grafana

[Website](#) | [GitHub](#) | [Portfolio](#)

- Architected a **4-node IoT platform** (ESP32, STM32 Black Pill, Arduino Nano, Pico 2W/MicroPython) collecting **18 real-time sensor streams** (vibration, IR, gas, RPM, moisture, sound); **FastAPI + PostgreSQL 15 + Redis 7** backend processing telemetry in under **200ms per request**
- **Isolation Forest anomaly detection** on every ingest, auto-generating maintenance tickets with diagnostic recommendations; STM32 runs **100Hz FFT vibration analysis** via ARM CMSIS-DSP (**9x speedup** over naive DFT)
- **JWT refresh flow, Google/GitHub OAuth, TOTP 2FA**, multi-tenant **RBAC**, rate limiting, immutable audit log; **WebSocket** live dashboard; **Prometheus + Grafana** monitoring; deployed on **DigitalOcean VPS + Nginx**

Two-Stage Audio Amplifier

2026

Analogue Design, PCB Design, KiCad, Proteus, Electronics

[Portfolio](#)

- Delivered a **full engineering workflow: simulation, breadboard validation, PCB layout and bench testing**
- Validated performance with measured output of **2.980 Vpp** against a **3 Vpp** design target

4x4x4 NeoPixel LED Cube

2025

Arduino, C++, WS2812B, Embedded Systems

[GitHub](#) | [Portfolio](#)

- Built a 64-LED interactive display with a **non-blocking state machine architecture** using millis()
- Introduced four animation modes, ambient-light adaptive brightness and debounced user controls

AstonCV - Full-Stack CV Database

2026

PHP, MySQL, CSS, Apache, Cloudflare

[Website](#) | [GitHub](#) | [Portfolio](#)

- Engineered **12+ security controls** including **bcrypt hashing, CSRF tokens, PDO prepared statements and brute-force protection**
- **Deployed live** with searchable profiles, sorting/filtering and server-side PDF export

CNC Milling Machine Control System

Arduino, C++, Safety Systems, State Machines

2024

[Portfolio](#)

- Engineered a **safety-focused controller** using **finite state machine logic** and strict transition validation
- Integrated door interlocks, **emergency stop handling**, timed cooldown and LCD status feedback for safe operation

git-unlocked - Open Source Git Course

Git, GitHub, GitLab, Markdown, Open Source

2026

[GitHub](#) | [Portfolio](#)

- Created a structured learning resource with **217+ topic files** covering fundamentals to advanced workflows across Git, GitHub, GitLab and 5 other platforms
- Maintained contributor workflow with issue templates, PR standards and CI checks

EXPERIENCE

Student Representative

Student advocacy, stakeholder communication, academic liaison

Aston Students' Union, Birmingham, UK

Sep 2025 - Present

- Collect and communicate student feedback to academic staff through **SSLC, Senate and Council meetings**
- Delivered improvements to learning delivery and the wider course experience

Engineering Operations & Maintenance Planning (Virtual Experience)

Maintenance planning, risk analysis, reporting

British Airways via Forage, Remote

Oct 2025

- Reviewed A320 maintenance schedules, diagnosed component faults and produced structured **WREQ engineering reports** with material forecasts and risk-mitigation recommendations for six aircraft

Smart Mobility & Digital Tech (Virtual Work Experience)

Intelligent transport systems, sensor integration, data-driven urban mobility

Yunex Traffic via Springpod, Remote

Aug 2025

- Examined **intelligent transport systems** and real-time pollution sensing workflows, evaluating digital solutions for their impact on air quality and urban mobility efficiency

Diplomatic, Consular and Estates Intern

Consular operations, diplomatic property, document processing, client support

Ghana High Commission, London, UK

Mar 2024 - Aug 2024

- Operated end-to-end consular processing workflows using official diplomatic management systems - handling application intake, **biometric data capture**, document authentication and physical visa and passport production from print to issuance, supporting **100+ daily client interactions**
- Coordinated **diplomatic property operations** by accompanying officers on site inspections and facility surveys, managing estates documentation, maintenance schedules and office workflows

Junior Apprentice HVAC Technician

Field servicing, diagnostics, installation support

Massive Refrigeration Services, Accra, Ghana

Jul 2019 - Jul 2021

- Serviced, installed and maintained **50+ air conditioning units** using standard diagnostic tools and field servicing techniques

RESEARCH & PUBLICATIONS

git-unlocked: An Open-Source Modular Curriculum for Git and Version Control Education

I. Adjei · Zenodo · Jun 2026

[Zenodo](#) | [DOI](#)

- Open-source modular curriculum covering **12 thematic sections and 217+ topic files** across Git, GitHub and GitLab; published as a citable research artefact with a permanent DOI

VOLUNTEERING

Student Judge - National Emerging Talent Awards 2026

targetjobs UK (Group GTI)

Feb 2026

- Evaluated employer submissions for **Best Placement or Internship Programme** using qualitative and scoring criteria; commended for a thorough, timely and high-quality approach

Fundraising Volunteer - 10 Days of 5K Challenge

Cancer Research UK

Feb - Mar 2026

- Completed **10 × 5km runs (50km+)** for the Cancer Research UK 10 Days of 5K Challenge, managing a fundraising page that contributed to a campaign raising over **£797,424.64** (plus £159,224.14 Gift Aid)

SPOKEN LANGUAGES

English (Full professional), **Twi & Ga** (Full Native), **French & Spanish** (Elementary)